UNITAT D'ESTADÍSTICA I BIOINFORMÀTICA

VHIR

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RedRedundance

A web app to summarize the results of functional enrichment analyses for one or multiple gene lists

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INTRODUCTION

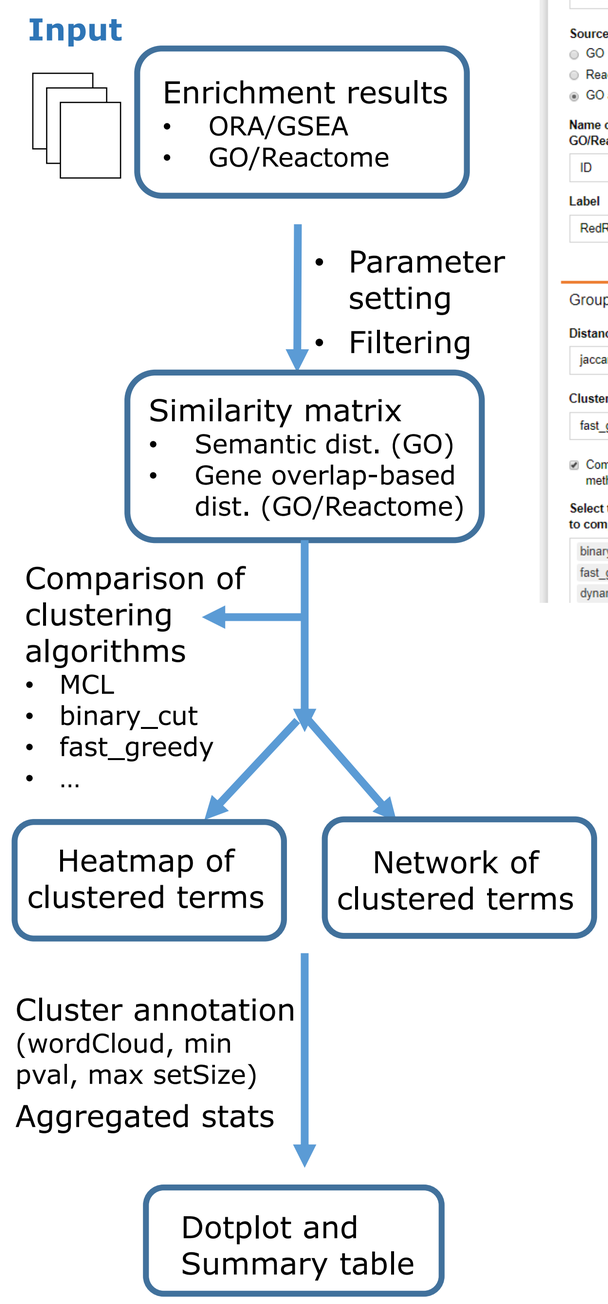
- Functional enrichment analysis allows identifying functional categories (e.g. Gene Ontology or pathway database annotations like KEGG or Reactome) that are statistically overrepresented among a list of features (i.e. genes).
- The interpretation of enrichment results is challenging due to the large amount of redundancy found between gene sets and across databases.
- *RedRedundance* aims at providing a user-friendly graphical interface that implements state-of-the-art software to summarize the results of enrichment analyses from individual gene sets into a more meaningful unit that brings meaning back to biological significance studies.

MAIN FEATURES

- ✓ **Interactive** R-Shiny App based on simplifyEnrichment [1] and igraph/visNetwork [2,3] R packages.
- ✓ Explore and **filter** the results of enrichment analyses
- ✓ Summarize and visually compare the results of enrichment analysis across **multiple lists and databases** (GO/Reactome). Many organisms available.
- ✓ Choose and compare among different algorithms for **clustering** individual gene sets into larger groups
- ✓ **Annotate** and **aggregate** cluster statistics
- ✓ Display results in **multiple visualization** options (Heatmaps, Networks, Dotplots, Tabular format…)

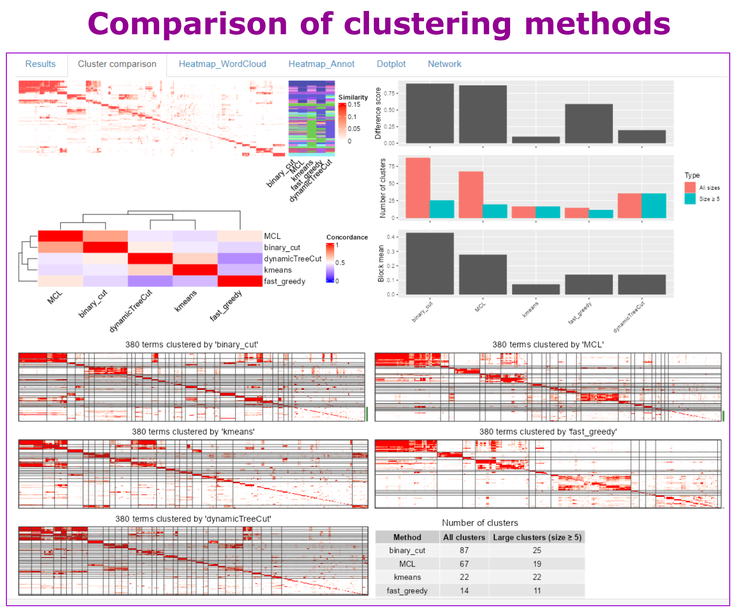
RESULTS

WORKFLOW

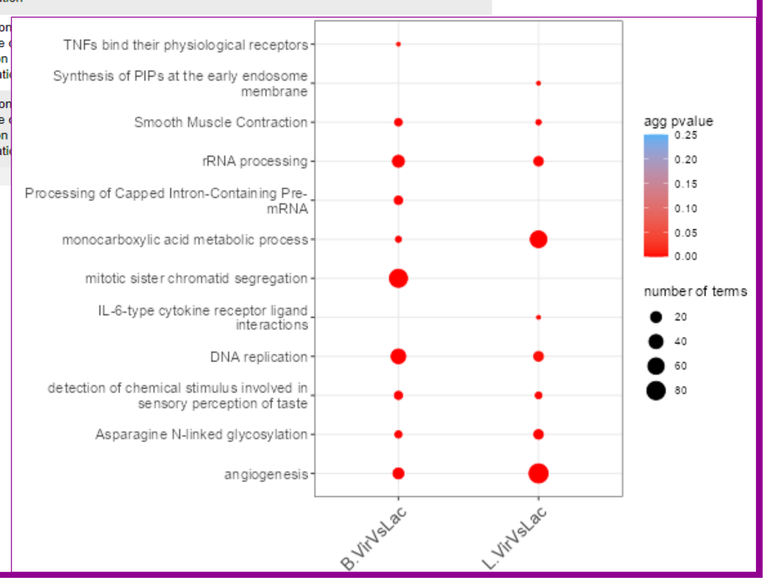


EXAMPLE OF OUTPUT

Comparison of clustering methods



Dotplot



- Example with Mouse mammary gland dataset (GSE60450)[4]. Gene Set Enrichment Analysis (GSEA) over GO.BP and ReactomePA databases was performed on preranked gene lists of Basal.VirginVsLactate and Lum.VirginVsLactate comparisons.

FUTURE WORK

- Extend to other databases
- Enhanced visualization options
- Implementation of different algorithms to aggregate per cluster statistics
- Interactive networks with collapsed information
- Cluster annotation
- Downloadable report

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AVAILABILITY

- The app can be freely accessed through the Web in: https://mferrer-almirall.shinyapps.io/RedRedundance/
- Or downloaded from Github in: https://github.com/mferrer-almirall/RedRedundance
- Contact: mireia.ferrer@vhir.org

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